Human Activity Recognition from Wi-Fi CSI Doppler Spectrograms: A CNN Baseline and a Contrastive Extension for Cross-Domain Generalization

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***Abstract*—We study device-free human activity recognition (HAR) from the Doppler spectrograms of Wi-Fi channel-state information (CSI), using the publicly released traces of the SHARP dataset [1], [2]. We first reproduce the Inception-style convolutional baseline of the course material (developed by M. Carraro): trained *only* on measurement set $S1$ and evaluated on the six unseen sets $S2$–$S7$, it reaches 48.2% accuracy on the unseen domains, quantifying how hard cross-domain generalization is when no domain-invariance mechanism is used. We then add a Supervised Contrastive Learning (SupCon) extension (developed by F. Vezzani): a deeper, batch-normalized encoder is pre-trained with a supervised-contrastive objective and domain-randomization augmentations, and a linear probe is fit on the frozen representation. This raises accuracy to 53.1%, improving over the baseline on *every* measurement set ($+4.9$ points overall), and a t-SNE analysis confirms the learned space is environment-invariant. Both models remain well below the full SHARP pipeline ($> 95\%$ on every set), whose strongest-path phase sanitization and multi-antenna *decision fusion* we do not implement; we analyse this gap and argue that the two ideas are complementary.**

***Index Terms*—Human Activity Recognition, Wi-Fi Sensing, Channel State Information, Convolutional Neural Networks, Contrastive Learning, Domain Generalization.**

I. Introduction

Contactless sensing of human activity is valuable in healthcare, security and ambient monitoring. Commodity Wi-Fi devices can be repurposed as radar-like sensors: as a person moves, their body induces micro-Doppler shifts on the Wi-Fi signal that are captured in the channel-state information (CSI) estimated by the receiver, *without recording any video*. After suitable processing, the CSI can be turned into a *Doppler spectrogram*—a time $\times$ velocity image—whose appearance changes markedly with the activity being performed, so the recognition problem can be cast as image classification with a convolutional neural network (CNN) [1].

The central difficulty is *generalization*. Different people move differently, and different rooms (and even the same room on different days) distort the signal differently, so a model that classifies perfectly in its training environment may collapse on a new one. This is the challenge raised in the project proposal, which explicitly suggests a *contrastive / self-supervised* approach to obtain subject- and environment-independent features. Prior systems close this gap mostly through signal processing—robust phase sanitization and multi-antenna fusion—at the cost of a heavy, hand-engineered pipeline [1]; we instead ask how far a purely *representation-learning* remedy can go on top of a minimal front-end.

In this report we tackle five-class Wi-Fi HAR under a strict cross-domain protocol (train on a single environment, test on six unseen ones) and study whether Supervised Contrastive Learning (SupCon) [3] can *learn* environment invariance rather than engineering it. The approach is data-efficient—it reuses the same small training set and a lightweight encoder—and its value is that the learned features can be reused by any downstream classifier and combined with the orthogonal signal-processing tools of the reference system. The report is organized around two clearly separated contributions:

- **Baseline (M. Carraro).** A faithful reimplementation, in plain PyTorch, of the dataset pipeline and of the baseline Inception-style CNN, with a cross-domain evaluation protocol (train on $S1$, test on $S2$–$S7$) and a per-measurement-set error breakdown.
- **Contrastive extension (F. Vezzani).** A Supervised Contrastive Learning stage (notebook §9): a deeper batch-normalized encoder is pre-trained with a supervised-contrastive loss and domain-randomization augmentations to learn environment-invariant features, then evaluated with a linear probe under the same protocol. It improves cross-domain accuracy over the baseline on every set.

We use the full SHARP system [1] as the reference upper bound and discuss, quantitatively, why both of our models sit below it and what the contrastive stage does and does not recover.

The remainder of the report is structured as follows. Section II reviews the related work on Wi-Fi sensing and contrastive representation learning. Section III gives a high-level view of the processing pipeline; Section IV details the signals and features, and Section V describes the two learning frameworks (baseline CNN and SupCon extension). The experimental results and the comparison against the SHARP

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Course project for Neural Networks and Deep Learning. Baseline pipeline by M. Carraro; Supervised Contrastive Learning extension by F. Vezzani.

paper are reported in Section VI, and Section VII draws the concluding remarks.

II. Related Work

Wi-Fi sensing and the SHARP system. Device-free HAR from commodity Wi-Fi has matured rapidly thanks to the availability of CSI extraction tools and public datasets. The most relevant work for us is SHARP [1], which recognizes human activities from the Doppler information contained in the CSI of IEEE 802.11ac links and, crucially, targets *environment and person independence*. SHARP achieves this through two ingredients: (i) a robust phase-sanitization step that isolates the multipath components and adopts the *strongest path* as the phase reference, and (ii) a *decision-fusion* scheme that runs a compact Inception-style CNN on each of the four receiver antennas and combines the per-antenna decisions. On the seven measurement sets of the accompanying CSI dataset [2], the full system exceeds 95% accuracy even on rooms, days and persons never seen at training time. These results are excellent, but they rest on a heavy, hand-engineered signal-processing front-end; the paper's own ablation shows that replacing the strongest-path sanitization with the standard reference-antenna method makes the *same* CNN collapse to 39–60% on the unseen sets, revealing how much of the robustness is carried by the pre-processing rather than by the learned representation. Our work keeps the minimal front-end (single antenna, no strongest-path sanitization) and asks how much cross-domain robustness can instead be recovered on the *representation* side.

Contrastive representation learning. Contrastive methods learn representations by pulling together "views" that should be similar and pushing apart those that should differ. SimCLR [4] established a simple self-supervised recipe: two random augmentations of the same image form a positive pair, all other images in the batch are negatives, and an encoder is trained with a temperature-scaled InfoNCE loss on L_2-normalized embeddings. Supervised Contrastive Learning (SupCon) [3] extends this idea to the labelled setting: *all* samples sharing the anchor's label become positives, which produces tighter, more class-consistent clusters than cross-entropy and improves robustness to natural corruptions. We adopt SupCon because its notion of "positive" maps naturally onto our goal: two windows of the same activity should be neighbours *regardless of the environment, person or antenna they come from*, so that the encoder is driven to discard domain-specific nuisances. To the best of our knowledge, combining SupCon with explicit domain-randomization augmentations for cross-domain Wi-Fi HAR on the SHARP data is the specific extension we contribute here: it complements the signal-processing route of [1] by attacking domain shift from the feature-learning side, and we quantify both its gains and its residual limits in Section VI.

III. Processing Pipeline

At a high level, both of our models cast Wi-Fi HAR as the classification of Doppler-spectrogram images and share the *same* data front-end; they differ only in *how* the representation is learned. Fig. 1 sketches the two chains. Common to both, the raw CSI Doppler traces are cut into fixed-size windows of shape $1 \times 340 \times 100$ (one antenna at a time), which become the network inputs (Section IV). From here the pipeline forks:

- the **baseline chain** (M. Carraro) feeds each window to an Inception-style CNN trained end-to-end with cross-entropy, and reads the activity off the softmax output;
- the **contrastive chain** (F. Vezzani) first applies domain-randomization augmentations to build two views of each window, pre-trains a deeper batch-normalized encoder with a Supervised Contrastive loss, then *freezes* the encoder and fits a linear probe on the resulting features.

Both chains are trained on a single environment ($S1$) and evaluated under the same cross-domain protocol (test on the unseen $S2$–$S7$), so that the two learning strategies can be compared on equal footing against the reference SHARP system [1]. The learning frameworks behind the two blocks are detailed in Section V.

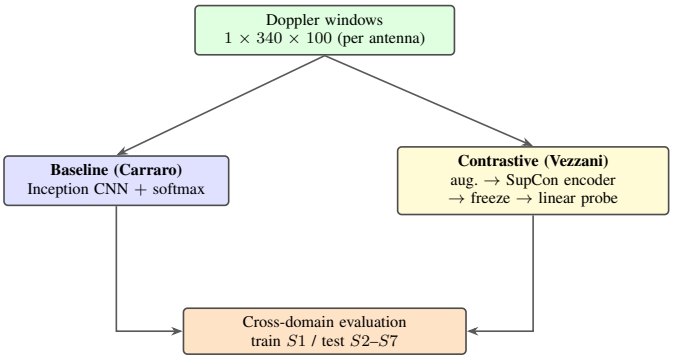


Fig. 1: High-level processing pipeline. A shared front-end turns CSI traces into Doppler windows; the baseline chain (M. Carraro) classifies them directly, while the contrastive chain (F. Vezzani) learns an environment-invariant representation before a linear probe. Both are assessed with the same cross-domain protocol.

IV. Signals and Features

(Dataset pipeline implemented by M. Carraro.)

A. The SHARP data

We use the pre-extracted Doppler traces released with the SHARP dataset [2], collected on IEEE 802.11ac, 80 MHz channels. Each trace is a pickled array of shape $(T, 100)$, where T is a variable number of time steps and 100 is the number of Doppler-velocity bins, and corresponds to a single receiving antenna (`stream_1...stream_4`) of a four-antenna monitor.

The recordings map to seven *measurement sets* $S1$–$S7$, each a distinct (environment, monitor position, person, day) combination (Tab. 1). Only $S1$ is used for training; the others probe generalization to unseen days ($S2$), unseen persons ($S3$), blocked line-of-sight ($S4, S5$), a different room ($S6$) and the worst case of simultaneously new room, day and person ($S7$).

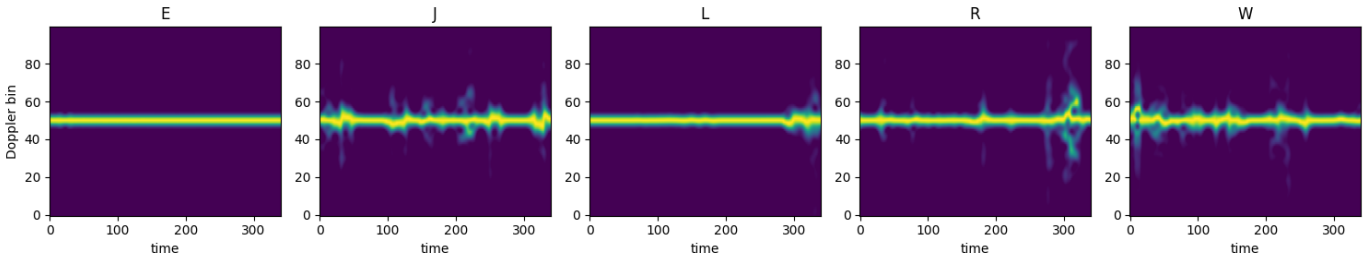


Fig. 2: One example Doppler-spectrogram window per activity (input to the network, transposed for readability: rows are the 100 Doppler-velocity bins, columns are the 340 time samples ≈ 2 s). Low-motion classes (*empty*, *sitting still*) concentrate energy around zero velocity, while *running/jumping* spread energy over a wide velocity range.

TABLE 1: Measurement sets of the SHARP dataset (after [1], Table 1). Mj: monitor position; Pi: person; LOS: direct path present.

Set	Monitor	Person	LOS	Role
$S1$	$M1$	$P1$	yes	**train/val/test**
$S2$	$M1$	$P1$	yes	test (new day)
$S3$	$M1$	$P2$	yes	test (new person)
$S4$	$M2$	$P1$	no	test (blocked LOS)
$S5$	$M2$	$P2$	no	test (blocked LOS)
$S6$	$M3$	$P1$	yes	test (new room)
$S7$	$M4$	$P3$	no	test (all new)

B. From traces to windows

To keep the setup comparable across *all* environments, we use the five core activities present everywhere: **E** (empty room), **J** (jumping), **L** (sitting still), **R** (running) and **W** (walking), matching the five-class task of the SHARP paper [1]. Each trace is cut into *non-overlapping* windows of $W_{\text{len}} = 340$ time samples (≈ 2 s); a channel dimension is added, giving inputs of shape $1 \times 340 \times 100$. Each of the four antenna streams is treated as an independent labelled sample: the baseline is therefore a *single-antenna* classifier that never fuses information across antennas—a point we return to in Section VI. This yields 13,564 windows, reasonably balanced across classes (E: 3012, J: 2160, L: 3056, R: 2684, W: 2652). Fig. 2 shows one window per activity.

C. Cross-domain split

Following the SHARP protocol, the split is by environment, never random: the folders of $S1$ ($S1a, S1b, S1c$) are used for training/validation and every other set is held out for testing; a random split would leak windows from the same person/room/day into both train and test and overestimate performance. Within $S1$ we keep 85% for training and 15% for validation (stratified), giving 2767 / 489 / 10,308 windows for train / val / test.

V. Learning Framework

A. Baseline CNN (M. Carraro)

The baseline reproduces the simplified reduced-Inception module of SHARP ([1], Fig. 4), shown in Fig. 3. Three parallel branches process the $1 \times 340 \times 100$ input and produce feature maps of identical spatial size 170×50 so they can be concatenated on the channel axis:

- **Branch 1:** $\text{MaxPool}(2 \times 2,\ \text{stride } 2) \rightarrow 1$ channel (no parameters);
- **Branch 2:** $\text{Conv}(5,\ 2 \times 2,\ \text{stride } 2) + \text{ReLU} \rightarrow 5$ channels;
- **Branch 3:** $\text{Conv}(3,\ 1 \times 1) \rightarrow \text{Conv}(6,\ 2 \times 2,\ \text{pad } 2) \rightarrow \text{Conv}(9,\ 4 \times 4,\ \text{stride } 2)$, each $+\,\text{ReLU} \rightarrow 9$ channels.

The branches are concatenated ($1{+}5{+}9 = 15$ channels), reduced to 3 channels by a 1×1 convolution, flattened, passed through $\text{Dropout}(0.2)$ and a final dense layer with one logit per class. The model has $\mathbf{128{,}535}$ trainable parameters—exactly the count reported for the SHARP baseline CNN [1].

Training uses cross-entropy loss and Adam (lr $= 10^{-3}$), mini-batches of 64, and *early stopping*: training halts when the validation loss has not improved for 7 epochs (budget 40), and the best (lowest validation-loss) weights are restored before testing. In our run, training stopped after epoch 13 (best validation loss 0.993). Fig. 4 shows the loss and accuracy curves: training accuracy keeps climbing while validation accuracy plateaus around 0.52—the classic signature of a model overfitting its single training environment.

B. Contrastive extension: Supervised Contrastive Learning (F. Vezzani)

(Developed by F. Vezzani; implemented and trained as Section 9 of the notebook.)

1) Motivation: The baseline is trained with plain cross-entropy on a single environment, so nothing pushes its features to be invariant across domains. Following the *contrastive / self-supervised* direction suggested in the project proposal, we implement *Supervised Contrastive Learning* (SupCon) [3]. The key change is *what* is optimized: instead of predicting the label directly, an *encoder* is first trained so that two windows of the *same activity* are pulled together and windows of different activities are pushed apart—*regardless of the environment, person or antenna they come from*. Because the objective depends only on activity identity, the encoder is encouraged to discard environment-specific nuisances. A tiny linear classifier is then fit on top of the frozen representation (*linear probe*).

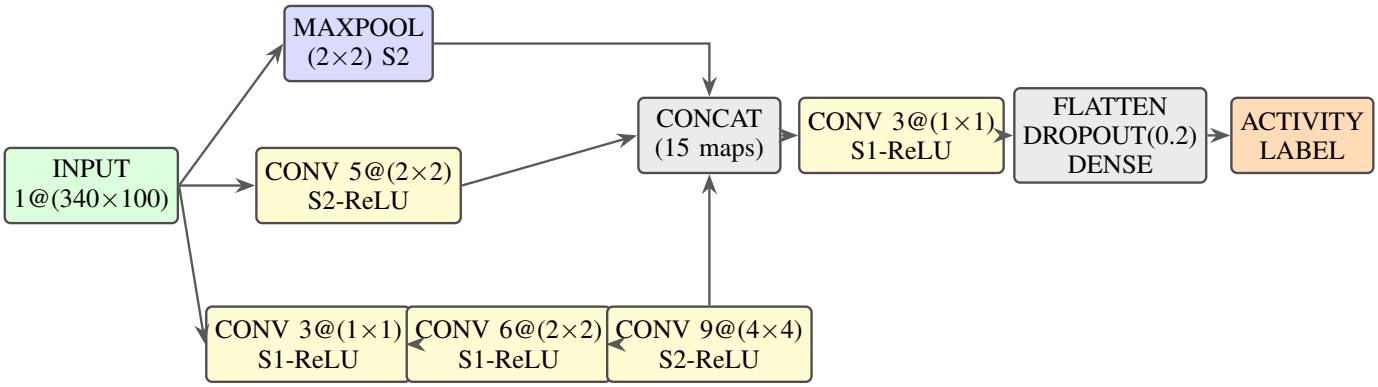


Fig. 3: Baseline single-antenna classification architecture (M. Carraro), a faithful reproduction of the SHARP reduced-Inception module ([1], Fig. 4). Three parallel branches (max-pool / strided conv / three stacked convs) extract multi-scale Doppler features; the maps are concatenated ($1{+}5{+}9 = 15$), reduced to 3 by a 1×1 convolution, then flattened and classified by a dense layer with dropout (128,535 parameters). S2 denotes stride 2; every CONV is followed by a ReLU.

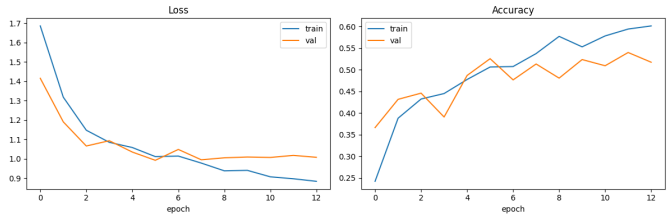


Fig. 4: Baseline training. Left: cross-entropy loss; right: accuracy. Validation accuracy plateaus near 0.52 while training accuracy keeps rising, i.e. the model overfits the single training set $S1$; early stopping halts training at epoch 13.

2) Domain-randomization augmentations: SupCon needs, for each window, two augmented "views" that keep the activity intact but perturb nuisance factors, acting as a light domain randomization (Fig. 5). All transforms run on-GPU on a whole batch (default parameters in parentheses):

- **amplitude scaling** by a per-window random gain ($\mathcal{U}(0.8, 1.2)$)—rooms/distances reflect the signal with different strength;
- **additive Gaussian noise** ($\sigma = 0.05$);
- **circular time shift** (± 40 samples)—the activity does not depend on where in the ≈ 2 s window it starts;
- **time and Doppler masking** (SpecAugment-style: a zeroed band of 40 time samples and 15 velocity bins)—forces the encoder to use the whole time–velocity pattern rather than any single region.

3) Encoder and projection head: The encoder $f(\cdot)$ is a deeper, batch-normalized evolution of the baseline's Inception block (Tab. 2). It keeps the three-branch Inception front-end (max-pool / strided conv / deeper conv, concatenated to 41 channels) but adds (i) batch normalization after every convolution, (ii) a 1×1 channel-mixing reduction to 32 channels, (iii) a second downsampling stage ($32 \to 64 \to 128$), and (iv) *global average pooling*, replacing the baseline's large `flatten`→dense layer with a fixed 128-dimensional representation h that has fewer parameters and is more shift-robust.

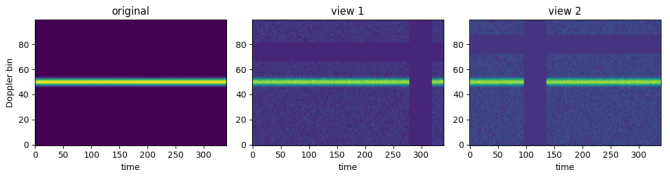


Fig. 5: A window (*left*) and two augmented views. Amplitude scaling and Gaussian noise change the texture; the circular time shift and the SpecAugment-style time/Doppler masks (dark bands) hide parts of the pattern, while the activity signature (the horizontal energy band) is preserved.

TABLE 2: Contrastive encoder $f(\cdot)$; output sizes for a $1 \times 340 \times 100$ input. BN+ReLU follow every convolution; "sN" denotes stride N (default stride 1). The pooling output is the 128-d representation h.

Stage	Layers	Output ($C \times H \times W$)
Inception	maxpool(2); conv(16,s2); conv(16)→conv(24,s2)	$41 \times 170 \times 50$
Reduce	conv(32, 1×1)	$32 \times 170 \times 50$
Stage 2	conv(64,s2)→conv(128,s2)	$128 \times 43 \times 13$
Pool	global average pool	128

A small MLP *projection head* $g(\cdot)$ ($128 \to 128 \to 128$, ReLU) maps h to an L_2-normalized embedding $\boldsymbol{z} = g(h)$ on the unit hypersphere; as prescribed by SupCon, the loss is applied to $\boldsymbol{z}$ and the head is *discarded* after pre-training, so classification uses h. The whole model has 130,952 parameters (encoder $\approx$97.9k + head $\approx$33k), comparable to the baseline's 128.5k.

4) SupCon loss: For a batch of L_2-normalized embeddings with labels, each anchor i attracts every other sample sharing its label (its positive set $P(i)$) and repels all others, through a temperature-scaled softmax over cosine similarities:

$$\mathcal{L} = \frac{1}{|I|} \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_p / \tau)}{\sum_{a \neq i} \exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_a / \tau)}, \qquad (1)$$

where I is the set of anchors with at least one same-class positive (the implementation averages the per-anchor terms

over the batch) and the temperature is $\tau = 0.07$. Unlike self-supervised SimCLR [4], using the labels in Eq. (1) makes *all* same-activity windows in the batch positives—even those from different files or environments—which turns generic contrastive learning into an environment-invariant *activity* representation.

5) Training and evaluation protocol: Everything uses the *same* $S1$-based split as the baseline.

- **Contrastive pre-training.** Each batch is expanded into two augmented views; encoder and head are trained jointly with Adam (lr 10^{-3}, weight decay 10^{-4}) for up to 30 epochs with early stopping (patience 7) on the validation contrastive loss, keeping the best encoder.
- **Linear probe.** The encoder is frozen, embeddings are pre-computed and standardized with training statistics, and a single $\mathrm{Linear}(128, 5)$ layer is trained with cross-entropy (Adam, lr 10^{-3}, early stopping). This isolates the quality of the representation.
- **Optional fine-tuning.** The encoder is then unfrozen and the full network fine-tuned end-to-end with a small learning rate (10^{-4}).

The model is evaluated under the same cross-domain protocol as the baseline (train on $S1$, test on $S2$–$S7$). A t-SNE projection of the test embeddings, colored by activity and by environment, is used as a diagnostic of environment-invariance.

VI. Results

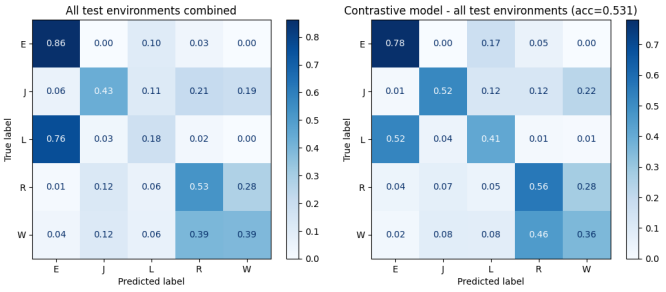
A. Baseline

On the pooled unseen test set the baseline reaches $\mathbf{48.2}\%$ accuracy (test loss 1.052); per set it ranges from 45.5% ($S2$) to 52.3% ($S7$) (Tab. 3). Its confusion matrix (Fig. 6 (a)) shows two dominant error modes: *sitting still* is absorbed by *empty* ($L \to E$: 0.76)—both are near-static and concentrate energy at zero Doppler velocity—and *running/walking* are heavily confused with each other ($W \to R$: 0.39, $R \to W$: 0.28).

B. Contrastive model

Fitting the linear probe on the frozen SupCon encoder raises overall accuracy to $\mathbf{53.1}\%$ (validation 58.3%), an improvement of $+4.9$ points over the baseline that holds on *every* measurement set (Tab. 3): the largest gains are on $S3$ ($+8.3$, unseen person) and $S7$ ($+5.9$, all-new). Its confusion matrix (Fig. 6 (b)) improves *running* recall markedly ($0.53 \to 0.56$ and, more importantly, fewer cross-class leaks) while the low-motion $L \to E$ confusion softens ($0.76 \to 0.52$).

The t-SNE projection (Fig. 7) is the most informative result. Colored by *environment* (right), the points are thoroughly mixed—the SupCon objective *did* achieve its design goal of an environment-invariant representation. Colored by *activity* (left), the space splits cleanly into a low-motion lobe (*empty*, *sitting*) and a high-motion lobe (*jumping*, *running*, *walking*), but the individual activities *within* each lobe remain partly entangled. Contrastive learning thus removed the *domain* nuisance but exposed a harder, intrinsic limit: a single-antenna ≈ 2 s Doppler window does not fully separate the activities,



(a) Baseline (b) Contrastive

Fig. 6: Row-normalized confusion matrices on all unseen test environments. The contrastive model (b) reduces the low-motion $L \to E$ confusion and sharpens *running* relative to the baseline (a).

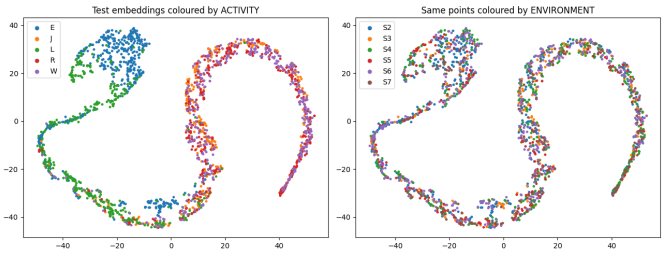


Fig. 7: t-SNE of the test embeddings learned by the SupCon encoder. *Left*, colored by activity: a low-motion lobe (*empty*, *sitting*) and a high-motion lobe (*jumping*, *running*, *walking*). *Right*, the same points colored by environment: the colors are mixed *inside* the structure, i.e. the representation is organized by *activity*, not by *environment*—the intended environment-invariance.

whichever way it is encoded. Consistently, the SupCon loss decreased only from ~ 4.6 to ~ 4.4 with a noisy validation curve—the pre-training did not fully converge.

Interestingly, *end-to-end fine-tuning* of the encoder *hurts*: it drops back to 48.4%, essentially the baseline level. Freezing the contrastively-pretrained encoder is what generalizes; fine-tuning lets the network re-overfit the single training environment. Two caveats temper the headline gain: the small baseline is high-variance across runs (a separate run reached $\sim 53\%$), so the $+4.9$-point improvement is best read as indicative and would ideally be confirmed over several seeds; and, as Tab. 3 shows, both models stay far below full SHARP.

C. Comparison with the SHARP paper

Tab. 3 places our results next to the SHARP paper [1]. Two SHARP columns are shown: the *full* system (strongest-path phase sanitization + 4-antenna decision fusion) and an *ablation* that replaces the strongest-path sanitization with the standard reference-antenna method, feeding the *same* CNN ([1], Table 4). Full SHARP stays above 95% on every unseen set. Our contrastive model (53.1%) reaches roughly the level of SHARP's single-component ablation ($\sim 53\%$ mean), and our baseline sits just below it. The ~ 45-point gap to full

TABLE 3: Per-set accuracy (%): training on $S1$, testing on unseen sets, all from one run. Ours: single-antenna baseline (M. Carraro) and SupCon linear-probe (F. Vezzani). SHARP-full and SHARP-ablation from [1] (Tables 3 and 4). Higher is better.

	Ours (this work)		**SHARP [1]**	
Set	Baseline (Carraro)	Contrastive (Vezzani)	Full (fusion)	Ablation (1 ant.)
$S2$	45.5	51.1	99.8	59.8
$S3$	51.4	59.7	99.3	58.0
$S4$	48.5	51.9	97.6	43.5
$S5$	47.5	52.9	96.6	39.2
$S6$	48.1	51.3	99.8	59.4
$S7$	52.3	58.2	96.0	60.3
Overall	**48.2**	**53.1**	$\sim$98	$\sim$53

SHARP is thus explained by the two ingredients we do not implement (Section VI-D), not by a bug in either model.

D. Discussion

a) Why both models are far from SHARP: Full SHARP owes its environment independence to two mechanisms we lack. *(i) Multi-antenna decision fusion:* SHARP runs the single-antenna classifier on each of the four antennas and combines the four decisions; input concatenation is markedly worse (e.g. $81.7\% \rightarrow 96.6\%$ on $S5$) and accuracy grows monotonically with the number of fused antennas [1], so our single-antenna model sits at the low end. *(ii) Robust phase sanitization:* SHARP uses the *strongest* multipath component as phase reference; its own ablation shows the standard reference-antenna method makes the *same* CNN collapse from $> 95\%$ to 39–60% (Tab. 3, last column)—right where our models land. Two further factors matter: we use *non-overlapping* windows (SHARP slides the window, yielding far more samples) and train on a single small environment, making the baseline high-variance.

b) What the contrastive extension achieves: The SupCon stage attacks domain shift from the *representation* side rather than the signal-processing side: it *learns* an embedding in which same-activity windows cluster across domains, aided by the domain-randomization augmentations. The t-SNE (Fig. 7) confirms this worked—the space is environment-invariant—and the linear probe improves accuracy on every set. Yet overall accuracy is still about half that of full SHARP, because the residual errors are *activity* confusions (walking vs. running, sitting vs. empty), not *domain* shift. The two ideas are thus *complementary*: contrastive learning handles the domain axis, while separating the activities most likely requires the multi-antenna information that SHARP's fusion exploits. Combining the frozen SupCon encoder (which generalizes far better than fine-tuning) with antenna fusion is the most promising direction left open.

VII. Concluding Remarks

We reproduced and evaluated a baseline Wi-Fi HAR pipeline under a strict cross-domain protocol (48.2% on unseen environments) and added a Supervised Contrastive Learning extension that improves cross-domain accuracy on every measurement set (53.1% overall, $+4.9$ points). A t-SNE analysis shows that the extension achieves its intended goal—an environment-invariant representation—while a direct comparison with the SHARP paper [1] places both models well below the full pipeline, whose multi-antenna decision fusion and strongest-path phase sanitization we do not implement.

The practical value of the extension is that it recovers domain robustness *on the learning side*, with a lightweight encoder, no extra signal-processing, and a frozen, reusable feature space. Since its residual errors are *activity* confusions rather than *domain* shift, the contrastive encoder is complementary to SHARP's antenna fusion: the clearest next step is to combine the two—keeping the frozen encoder—ideally with a sliding-window dataset and multi-seed runs.

a) What we learned and difficulties encountered: The main lesson was that a strict, environment-based split changes the story: modest-looking numbers ($\sim 50\%$) are the honest cost of never seeing the test environment, which a random split would have hidden. The t-SNE proved valuable as a diagnostic—it revealed that contrastive learning had solved the *domain* axis but not the *activity* axis. The main difficulties were: (i) getting SupCon pre-training to converge (the loss barely moved), so the representation is useful but under-trained; (ii) the counter-intuitive finding that end-to-end fine-tuning *hurt* generalization by re-overfitting the single environment, which taught us to trust the frozen-encoder / linear-probe evaluation; and (iii) the high run-to-run variance of the small baseline, which makes single-run comparisons fragile and calls for multi-seed averaging.

Author Contributions

Marco Carraro: dataset/windowing pipeline, baseline Inception CNN, training with early stopping, and the cross-domain evaluation (Section IV and the baseline parts of Section V–VI). **Francesco Vezzani:** the Supervised Contrastive Learning extension (notebook §9)—augmentations, the batch-normalized encoder with global average pooling, the projection head and SupCon loss, contrastive pre-training, linear-probe / fine-tuning evaluation, and the t-SNE analysis (the contrastive parts of Section V–VI).

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